

ABSTRACT

Purpose: The project is developed to drive an induction motor for the application of forward and reverse direction as well as to control the speed using wireless technology. For example, an exhaust fan can be used in both the direction to fresh air in and throw hot air out. This can be used in the case of conventional exhaust fan that rotates in one direction only.

Design and Description: This system is equipped with induction motor speed controlling system. Each time data sent by IR Remote, as per the code written and is executed by the micro controller to deliver delayed firing pulse of Thyristor. There by the speed of induction motor can be controlled. The running speed of motor is displayed by LCD Module.

Methodology: By using IR remote method, the speed and direction of induction motor should be controlled.

Findings: This proposed system demonstrates a technology to rotate a induction motor in both clockwise and counter clockwise direction.

Keywords: 2x16 LCD module, relay, AVR microcontroller, 12-0-12 Transformer , IR remote with Receiver, Induction Motor, Electronic components BT-136 TRAIC, MCT2E and MOC3011 - Optical Isolator, other electronic components.

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Chapter-1

Introduction

Generally, a Bidirectional rotation of an induction motor with a remote control is a device that used to control a single phase ac motor rotation in the directions of clockwise and anti-clockwise. In straightway, we cannot turn the single-phase ac motor in both directions as well as we can only control this motor with a single phase one way switch. In other words, if we want to manage the motor in both directions then we must use any kind of motor drive, which could be so expensive, but it would also function with single phase one way switch. On concentrating these problems, here we have implemented a device that is named as a bidirectional rotation of an induction motor with a remote control device.

The electrical direction control has many economical as well as engineering advantages over mechanical direction control. Historically any industrial equipment have its speed change mechanically by means of stepped pulleys or by the gear mechanism. With this proposed system, simply with the command inputs which are given to the input terminals validates and perform three modes of operations in accordance with the validated output of the controller. And initially controller is being dumped with input commands inform of hexa-code. Hence the controller executes the load to rotate forward and reverse directions. And we are supposed to implement this technique on dc motor. Further it can extended to ac motors also which has a wide range of applications

1.1 Induction Motor

An induction or asynchronous motor is an AC electric motor in which the electric current in the rotor needed to produce torque is obtained by electromagnetic induction from the magnetic field of the stator winding. An induction motor therefore does not require mechanical commutation, separate-excitation or self-excitation for all or part of the energy transferred from stator to rotor, as in universal, DC and large synchronous motors. An induction motor's rotor can be either wound type or squirrel-cage type.

1.2 Working of Induction Motor

Single phase induction motors require just one power phase for their operation. They are commonly used in low power rating applications, in domestic as well as industrial use.

This article is aimed at giving you a conceptual overview of the working of single phasemotors. The following article gives a detailed description of the video lecture.

1.3 Parts of a Single Phase Induction Motor

The main components of a single phase motor are the rotor and stator winding. The rotor is the rotating part, the stator winding helps in rotating the rotor. In Fig.1 the iron layer lamina which is fitted inside rotor is not shown, for better viewing of the rotor bars.

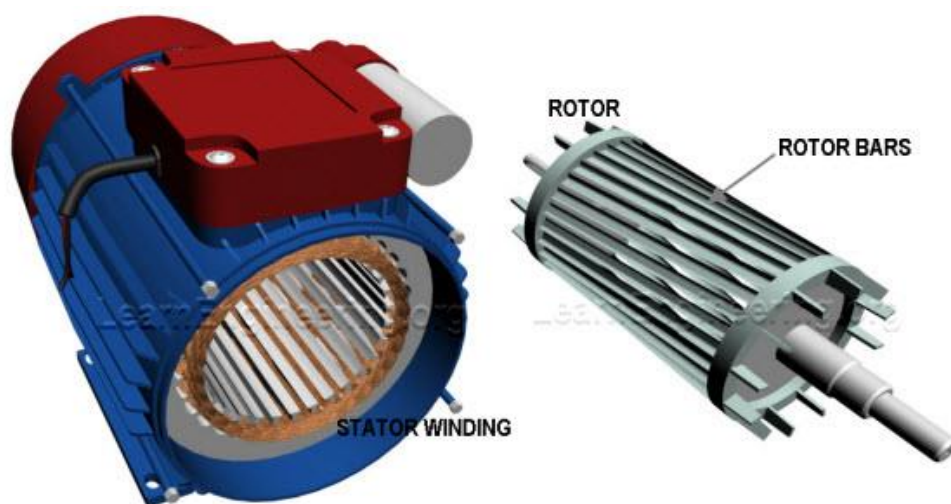


Fig.1.1 Mains parts of single phase induction motor : Rotor and Stator

The winding has got 2 parts one main winding and an auxiliary winding. The auxiliary winding is placed perpendicular to the main winding. A capacitor is connected to the auxiliary winding.

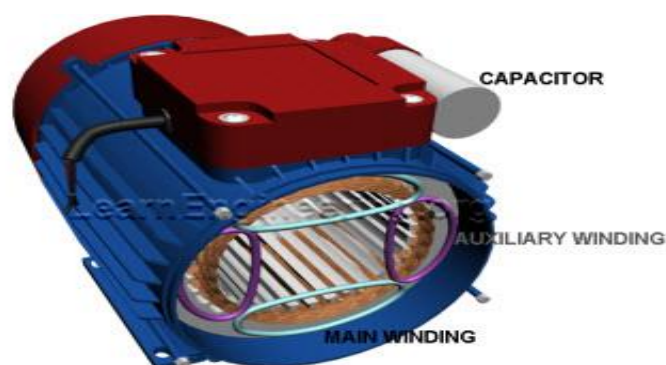


Fig.1.2 Single phase motor has got 2 perpendicularly placed windings

1.4 Advantages of Bidirectional Rotation of an Induction Motor with a Remote Control Device

- By using this bidirectional rotation of an induction motor with a remote control device the user can easily rotate the split phase induction motor in either direction.
- This bidirectional rotation of an induction motor device could be easily used with domestic and industrial exhaust fans for fresh air in and hot air throw out.
- By using bidirectional rotation of an induction motor device, the user can easily on or off the split phase inducing motor with some distance without wasting any time.

Chapter- 2

Literature review

Saidur,R, Mekhilef,S Ali, M. A. Safari.A . Mohammed.H.A proposed Applications of variable speed drive (VSD) in electrical motors, Renewable and Sustainable Energy. Most motors are designed to operate at a constant speed and provide a constant output; however, modern technology requires different speeds in many applications where electric motors are used. A variable speed drive (VSD) is a device that regulates the speed and rotational force, or output torque of mechanical equipment. Effects of applying VSDs are in both productivity improvements and energy savings in pumps, fans, compressors and other equipment. Ibrahim.M Alsofyani, N. R. Idris.N proposed Variable frequency drives (VFDs) can provide reliable dynamic systems and significant savings in energy usage and costs of the induction motors (IMs). Sensorless controlled IM drives have advantages in terms of efficiency enhancement and energy savings for critical applications such as electric vehicles, high performance machine tools, fans, compressors, etc. IM drives without having speed sensors or optical encoders mounted at the motor shaft are attractive because of their lower cost and higher reliability.

Nabae.A Kenichi.O Hiroshi.U. Kurosawa,.R-Induction motors can be controlled quickly and stably by controlling the stator current as a vector quantity with excitation and torque components. There are two such control methods: 1) the magnetic flux vector feedback method, invented by Blaschke, , where the magnetic flux vector from the induction motor is fed back and the stator current vector is controlled on the basis of the flux vector; and 2) the slip frequency control method ,where the angular velocity of the rotor is fed back, and the stator current vector is controlled on the basis of the velocity.

Bose.B.K proposed Power electronics and drives. Power electronics is an area of extremely important and rapidly changing technology. Technological advancements in the area contribute to performance improvement and cost reduction, with applications proliferating in industrial, commercial, residential, military and aerospace environments. This book is meant to help engineers operating in all these areas to stay up-to-date on the most recent advances in the field, as well as to be a vehicle for clarifying increasingly complex theories and mathematics. This book will be a cost-effective and convenient way for engineers to get up-to-speed on the latest trends in power electronics.

Stanley.H.C proposed analysis of induction motor. Analysis studies of an asynchronous motor with a tapered geometry for a two-person electric vehicle were carried out. The goal of

this study is to analyze change in the taper angle for this asynchronous motor in terms of the power factor, efficiency, starting torque, breakdown torque, and rated torque. First, the vehicle dynamics and the asynchronous motor structure were analyzed. Then, axial, radial, and hybrid flux electric motor models and the purpose of these models were investigated. Literature studies have shown that there is no work related to angle optimization for a flux with a conical geometry hybrid structure for an asynchronous motor.

Induction machines are by far the largest group of all industrial electrical machines, converting approximately 70-80 % of all electrical energy into mechanical form. It has a very robust rotor construction, which makes it suitable for high-speed applications, and further, with a proper design, it can have good overloading and field weakening characteristics. The theory of induction machines is old and well-known, and therefore, both motors and inverters are widely available from numerous manufacturers from fractional kW machines up to MW range. The induction motor is also known as the asynchronous motor, which derives from the fact that the rotor is always lagging the stator magnetic field. The difference is called the slip, and it is a fundamental characteristic in the operation of an induction motor. The slip is problematic in drives where a high dynamic performance is required, as it degrades the transient response of the motor for instance during stepwise loading variations. Also the rotor copper losses are directly proportional to the slip. The slip can be decreased by reducing the rotor resistance, and also by using a higher air gap flux density. The biggest drawback of the induction machine is the always lagging power factor, because the machine is magnetized from the stator, in other words, there is a magnetizing current flowing in the stator winding even at no-load conditions. This means that less torque is available with a given current than for example with a PMSM, or alternatively, more current is required to produce an equal torque, which leads to an inverter with a higher current rating. With four-pole industrial induction machines, the power factor typically varies between 0.8-0.9, but with low-power induction machines, it can be notably low.

Chapter-3

Components Description

3.1 Atmega 32 Microcontroller

The AVR is a modified Harvard architecture 8-bit RISC single-chip microcontroller, which was developed by Atmel in 1996. The AVR was one of the first microcontroller families to use on-chip flash memory for program storage, as opposed to one-time programmable ROM, EPROM, or EEPROM used by other microcontrollers at the time. megaAVR chips became popular after they were designed into the 8-bit Arduino platform.

Atmega 32 is an 8-bit high performance microcontroller of Atmel's Mega AVR family with low power consumption. Atmega 32 is based on enhanced RISC (Reduced Instruction Set Computing, Know more about RISC and CISC Architecture) architecture with 131 powerful instructions. Most of the instructions execute in one machine cycle. Atmega 32 can work on a maximum frequency of 16MHz.

Atmega 32 has 16 KB programmable flash memory, static RAM of 1 KB and EEPROM of 512 Bytes. The endurance cycle of flash memory and EEPROM is 10,000 and 100,000, respectively.

Atmega 32 is a 40 pin microcontroller. There are 32 I/O (input/output) lines which are divided into four 8-bit ports designated as PORTA, PORTB, PORTC and PORTD.

Atmega 32 has various in-built peripherals like USART, ADC, Analog Comparator, SPI, JTAG etc. Each I/O pin has an alternative task related to in-built peripherals. The following table shows the pin description of Atmega 32.

3.1.1 Pin Configuration of Atmega 32

Table: IPin Configuration of Atmega 32

Pin No.	Pin name	Description	Alternate Function
1	(XCK/T0) PB0	I/O PORTB, Pin 0	T0: Timer0 External Counter Input. XCK : USART External Clock I/O
2	(T1) PB1	I/O PORTB, Pin 1	T1: Timer1 External Counter Input
3	(INT2/AIN0) PB2	I/O PORTB, Pin 2	AIN0: Analog Comparator Positive I/P INT2: External Interrupt 2 Input
4	(OC0/AIN1) PB3	I/O PORTB, Pin 3	AIN1: Analog Comparator Negative I/P OC0 : Timer0 Output Compare Match Output
5	(SS) PB4	I/O PORTB, Pin 4	In System Programmer (ISP) Serial Peripheral Interface (SPI)
6	(MOSI) PB5	I/O PORTB, Pin 5	
7	(MISO) PB6	I/O PORTB, Pin 6	
8	(SCK) PB7	I/O PORTB, Pin 7	
9	RESET	Reset Pin, Active Low Reset	
10	V _{cc}	V _{cc} = +5V	
11	GND	GROUND	
12	XTAL2	Output to Inverting Oscillator Amplifier	
13	XTAL1	Input to Inverting Oscillator Amplifier	
14	(RXD) PD0	I/O PORTD, Pin 0	USART Serial Communication Interface
15	(TXD) PD1	I/O PORTD, Pin 1	
16	(INT0) PD2	I/O PORTD, Pin 2	External Interrupt INT0
17	(INT1) PD3	I/O PORTD, Pin 3	External Interrupt INT1
18	(OC1B) PD4	I/O PORTD, Pin 4	PWM Channel Outputs
19	(OC1A) PD5	I/O PORTD, Pin 5	
20	(ICP) PD6	I/O PORTD, Pin 6	Timer/Counter1 Input Capture Pin
21	PD7 (OC2)	I/O PORTD, Pin 7	Timer/Counter2 Output Compare Match

			Output
22	PC0 (SCL)	I/O PORTC, Pin 0	TWI Interface
23	PC1 (SDA)	I/O PORTC, Pin 1	
24	PC2 (TCK)	I/O PORTC, Pin 2	JTAG Interface
25	PC3 (TMS)	I/O PORTC, Pin 3	
26	PC4 (TDO)	I/O PORTC, Pin 4	
27	PC5 (TDI)	I/O PORTC, Pin 5	
28	PC6 (TOSC1)	I/O PORTC, Pin 6	Timer Oscillator Pin 1
29	PC7 (TOSC2)	I/O PORTC, Pin 7	Timer Oscillator Pin 2
30	AVcc	Voltage Supply = Vcc for ADC	
31	GND	GROUND	
32	AREF	Analog Reference Pin for ADC	
33	PA7 (ADC7)	I/O PORTA, Pin 7	ADC Channel 7
34	PA6 (ADC6)	I/O PORTA, Pin 6	ADC Channel 6
35	PA5 (ADC5)	I/O PORTA, Pin 5	ADC Channel 5
36	PA4 (ADC4)	I/O PORTA, Pin 4	ADC Channel 4
37	PA3 (ADC3)	I/O PORTA, Pin 3	ADC Channel 3
38	PA2 (ADC2)	I/O PORTA, Pin 2	ADC Channel 2
39	PA1 (ADC1)	I/O PORTA, Pin 1	ADC Channel 1
40	PA0 (ADC0)	I/O PORTA, Pin 0	ADC Channel 0

3.1.2 Features of AVR Microcontroller

- Multifunction, bi-directional general-purpose I/O ports with configurable, built-in pull-up resistors
- Multiple internal oscillators, including RC oscillator without external parts
- Internal, self-programmable instruction flash memory up to 256 kB (384 kB on XMega)
- On-chip debugging (OCD) support through JTAG or debug WIRE on most devices
- Internal data EEPROM up to 4 kB
- Internal SRAM up to 32 kB (32 kB on XMega)
- External 64 kB little endian data space on certain models, including the Mega8515 and Mega162.
- 8-bit and 16-bit timers
- Analog comparator
- 10 or 12-bit A/D converters, with multiplex of up to 16 channels
- 12-bit D/A converters
- A variety of serial interfaces, including
- Brownout detection
- Watchdog timer (WDT)
- Multiple power-saving sleep modes
- Lighting and motor control (PWM-specific) controller models
- CAN controller support
- USB controller support
- Ethernet controller support
- LCD controller support
- Low-voltage devices operating down to 1.8 V (to 0.7 V for parts with built-in DC–DC upconverter)
- picoPower devices
- DMA controllers and "event system" peripheral communication.
- Fast cryptography support for AES and DES

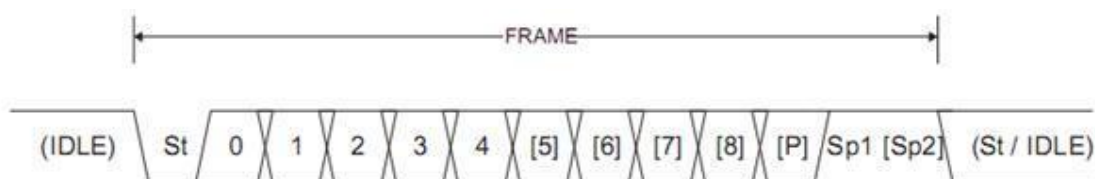
3.2 Usart

Communication between two entities is important for the information flow to take place. In general, the information transport system can be parallel in which the complete byte of data is sent at a time, with each bit having a separate dedicated line or it can be serial where only one communication line is available which is shared by all the bits sequentially. The pros and cons of these two systems are equivalent and selection between the two depends on the application.

Data can be exchanged using parallel or serial techniques. Setup for parallel data transfer is not cost effective but is a very fast method of communication. Serial communication is cost effective because it requires only a single line of connection but on the other hand is a slow process in comparison to parallel communication. This article explains serial communication of AVR microcontroller (Atmega 32) with PC. The data is transmitted from the controller using RS232 standard and displayed on the PC using Hyper Terminal.

There are two methods for serial data communication (i) Synchronous and (ii) Asynchronous communication. In Synchronous communication method complete block (characters) is sent at a time. It doesn't require any additional bits (start, stop or parity) to be added for the synchronization of frame. The devices are synchronized by clock. And in asynchronous communication data transmission is done byte by byte i.e., one byte at a time. The additional bits are added to complete a frame.

In synchronous communication the frame consists of data bits while in asynchronous communication the total number of bits in a frame may be more than the data bits.



There are three ways in which serial communication can be done



Fig.3.1 Simplex communication

Transmission is done in one direction.



Fig.3.2 Half duplex communication

Transmission can be done in both the direction but one side at a time.

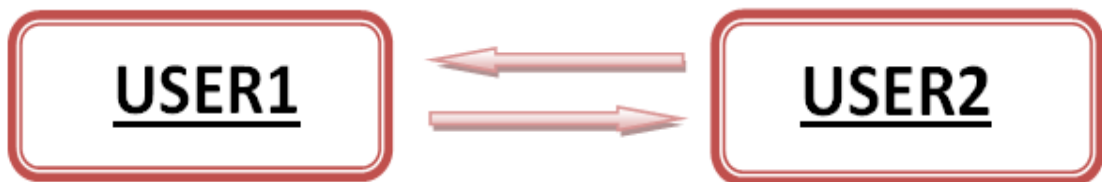


Fig.3.3 Full duplex communication

Atmega 32 is equipped with three different kinds of serial communication peripheral systems.

- i. Serial USART
- ii. SPI (Serial Peripheral Interface)
- iii. TWI (Two wire Interface)

SERIAL USART (universal synchronous asynchronous receiver and transmission/ transmitter):

Serial USART provides full-duplex communication between the transmitter and receiver. Atmega 32 is equipped with independent hardware for serial USART communication. Pin-14 (RXD) and Pin-15 (TXD) provide receive and transmit interface to the microcontroller.

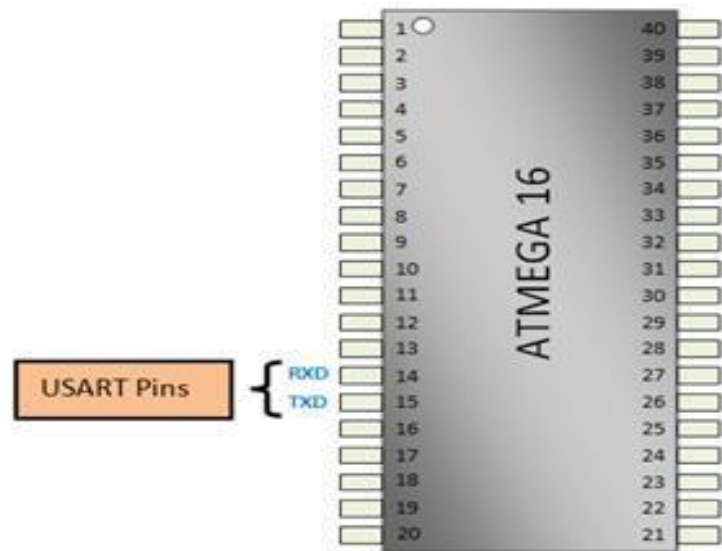


Fig:3.4Usart pins

Atmega 32 USART provides asynchronous mode of communication and do not have a dedicated clock line between the transmitting and receiving end. The synchronization is achieved by properly setting the baud rate, start and stop bits in a transmission sequence.

Start bit and stop bit: These bits are use to synchronize the data frame. Start bit is one single low bit and is always given at the starting of the frame, indicating the next bits are data bits. Stop bit can be one or two high bits at the end of frame, indicating the completion of frame.

One frame=start bit + data bit +stop bit

Baud Rate: In simple words baud rate is the rate at which serial data is being transferred.

Atmega 32 USART has following features:

- Different Baud Rates.
- Variable data size with options ranging from 5bits to 9bits.
- One or two stop bits.
- Hardware generated parity check.
- USART can be configured to operate in synchronous mode.
- Three separate interrupts for RX Complete, TX complete and TX data register empty.

USART Registers

To use the USART of Atmega 32, certain registers need to be configured.

UCSR: USART control and status register. It's is basically divided into three parts UCSRA, UCSRB and UCSRC. These registers are basically used to configure the USART.

UBRR: USART Baud Rate Registers. Basically use to set the baud rate of USART

UDR: USART data register

3.3 Analog To Digital Converter

Microcontroller understands only digital language. However, the inputs available from the environment to the microcontroller are mostly analog in nature, i.e., they vary continuously with time. In order to understand the inputs by the digital processor, a device called Analog to Digital Converter (ADC) is used. As the name suggests this peripheral gathers the analog information supplied from the environment and converts it to the controller understandable digital format, microcontroller then processes the information and provides the desired result at the output end.

Atmega 32 has an inbuilt 10 bit, 8-channel ADC system. Some of the basic features of Armega16 ADC are:

- 8 Channels.
- 10-bit Resolution.
- Input voltage range of 0 to Vcc.
- Selectable 2.56V of internal Reference voltage source.
- AREF pin for External Reference voltage.
- ADC Conversion Complete Interrupt.

ADC channels in Atmega 32 are multiplexed with PORTA and use the common pins (pin33 to pin40) with PORTA. ADC system of Atmega 32 microcontroller consists of following pins:

- i. **ADC0-ADC7:** 8 Channels from Pin 40 to Pin 33 of Atmega 32 ADC peripheral.
- ii. **AREF:** Pin32 of Atmega 32 microcontroller, the voltage on AREF pin acts as the reference voltage for ADC conversion, reference voltage is always less than or equal to the supply voltage, i.e., Vcc.

iii.**AVCC:** Pin30, this pin is the supply voltage pin for using PORTA and the ADC; AVCC pin must be connected to Vcc (microcontroller supply voltage) to use PORTA and ADC.

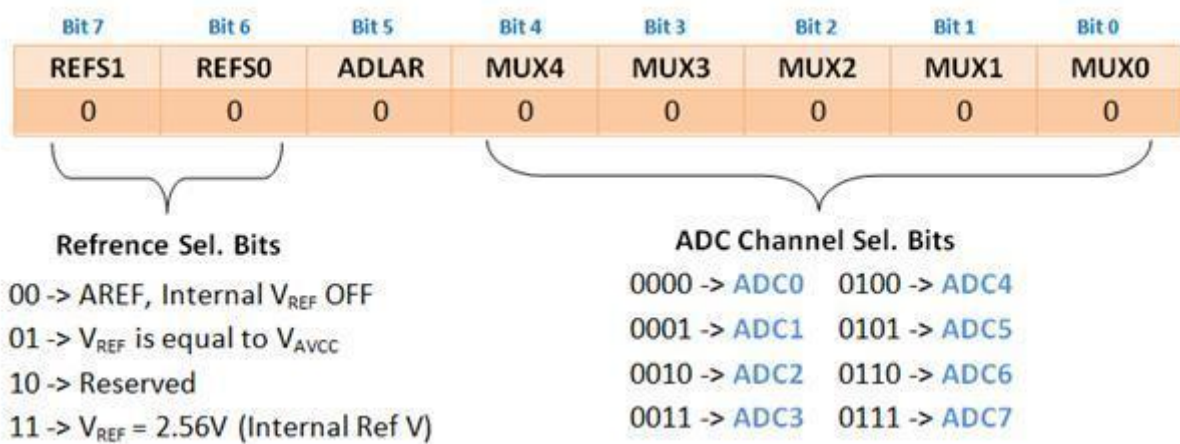
Note: External reference voltage source can be used at AREF pin. However, Atmega 32 also has internal reference voltage options of 2.56V and $V_{ref} = V_{cc}$.

The figure below shows the pin configuration for ADC system of Atmega 32 microcontroller.

3.3.1 ADC Registers

To use the ADC peripheral of Atmega 32, certain registers need to be configured.

i. **ADMUX** (ADC Multiplexer And Selection Register)



REFS[0:1] bits determine the source of reference voltage whether it is internal or the external voltage source connected to AREF pin. **MUX[4:0]** bits are used to select between the channels which will provide data to ADC for conversion. **ADLAR** bit when set to 1 gives the left adjusted result in data registers ADCH and ADCL.

ii.**ADCSRA** (ADC Control and Status Register)

ADEN: ADC Enable bit, this bit must be set to 1 for turning ADC on.

ADSC: ADC Start Conversion bit, this bit is set to 1 to start ADC conversion, as soon as conversion is completed this bit is set back to 0 by the hardware.

ADATE: ADC Auto Trigger Enable, this bit is set to 1 to enable auto triggering of ADC conversion.

ADIF: ADC Interrupt Flag, this bit is set to 1 when ADC conversion gets complete.

ADIE: ADC Interrupt Enable, this bit is set to 1 if we want to activate the ADC conversion complete interrupt.

ADPS[0:2]: ADC Prescaler bits, these bits are used to set the ADC clock frequency, the configuration of these bits determine the division factor by which the microcontroller clock frequency is divided to get the ADC clock frequency. The figure above shows the prescaler bit values for respective division factor.

ADC Clock Frequency = XTAL Frequency/Prescaler

Example: XTAL Freq = 12MHz

Prescaler = 32

ADC Clock Freq = $12000000/32$

=375KHz

The ADC clock frequency must lie somewhere between 50 KHz to 200 KHz.

iii. **ADCH & ADCL** (ADC Data Registers)

When the ADC conversion is complete the data is stored in these two registers. The data configuration depends on the ADLAR bit value of ADMUX register. If ADLAR=0, data is right adjusted and if ADLAR=1, data is left adjusted. Always read ADCL first and then ADCH. In cases where the 8-bit precision is enough set the ADLAR bit to 1 to left adjust the data and read only the ADCH data register.

When ADLAR = 0,

ADCH	-	-	-	-	-	-	ADC9	ADC8
ADCL	ADC7	ADC6	ADC5	ADC4	ADC3	ADC2	ADC1	ADC0

When ADLAR = 1,

ADCH	ADC9	ADC8	ADC7	ADC6	ADC5	ADC4	ADC3	ADC2
ADCL	ADC1	ADC0	-	-	-	-	-	-

3.4LCD 2X16 Module

A liquid-crystal display (LCD) is a flat-panel display or other electronic visual display that uses the light-modulating properties of liquid crystals. Liquid crystals do not emit light directly. LCDs are used in a wide range of applications including computer monitors, televisions, instrument panels, aircraft cockpit displays, and signage. They are common in consumer devices such as DVD players, gaming devices, clocks, watches, calculators, and telephones, and have replaced cathode ray tube (CRT) displays in nearly all applications. They are available in a wider range of screen sizes than CRT and plasma displays, and since they do not use phosphors, they do not suffer image burn-in. LCDs are, however, susceptible to image persistence.

The LCD screen is more energy-efficient and can be disposed of more safely than a CRT. Its low electrical power consumption enables it to be used in battery-powered electronic equipment more efficiently than CRTs. It is an electronically modulated optical device made up of any number of segments controlling a layer of liquid crystals and arrayed in front of a light source (backlight) or reflector to produce images in color or monochrome.

LCD (Liquid Crystal Display) screen is an electronic display module and find a wide range of applications. A 16x2 LCD display is very basic module and is very commonly used in various devices and circuits. These modules are preferred over seven segments and other multi segment LEDs. The reasons being: LCDs are economical; easily programmable; have no limitation of displaying special & even custom characters (unlike in seven segments), animations and so on.

A 16x2 LCD means it can display 16 characters per line and there are 2 such lines. In this LCD each character is displayed in 5x7 pixel matrix. This LCD has two registers, namely, Command and Data.

The command register stores the command instructions given to the LCD. A command is an instruction given to LCD to do a predefined task like initializing it, clearing

its screen, setting the cursor position, controlling display etc. The data register stores the data to be displayed on the LCD. The data is the ASCII value of the character to be displayed on the LCD.

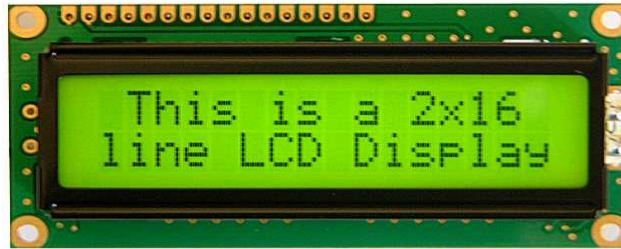


Fig.3.5LCD 2X16 Module

Pin DescriptionLCD 2X16 Module

Table:2pin Description LCD 2X16 module

Pin No	Function	Name
1	Ground (0V)	Ground
2	Supply voltage; 5V (4.7V – 5.3V)	V _{CC}
3	Contrast adjustment; through a variable resistor	V _{EE}
4	Selects command register when low; and data register when high	Register Select
5	Low to write to the register; High to read from the register	Read/write
6	Sends data to data pins when a high to low pulse is given	Enable
7	8-bit data pins	DB0
8		DB1
9		DB2
10		DB3
11		DB4

12		DB5
13		DB6
14		DB7
15	Backlight V _{CC} (5V)	Led+
16	Backlight Ground (0V)	Led-

VEE pin is meant for adjusting the contrast of the LCD display and the contrast can be adjusted by varying the voltage at this pin. This is done by connecting one end of a POT to the V_{CC} (5V), other end to the Ground and connecting the center terminal (wiper) of the POT to the VEE pin.

The JHD162A has two built in registers namely data register and command register. Data register is for placing the data to be displayed, and the command register is to place the commands. The 16×2 LCD module has a set of commands each meant for doing a particular job with the display.

High logic at the RS pin will select the data register and Low logic at the RS pin will select the command register. If we make the RS pin high and put a data in the 8 bit data line (DB0 to DB7), the LCD module will recognize it as a data to be displayed. If we make RS pin low and put a data on the data line, the module will recognize it as a command.

R/W pin is meant for selecting between read and write modes. High level at this pin enables read mode and low level at this pin enables write mode. E pin is for enabling the module. A high to low transition at this pin will enable the module.

DB0 to DB7 are the data pins. The data to be displayed and the command instructions are placed on these pins. LED+ is the anode of the back-light LED and this pin must be connected to V_{CC} through a suitable series current limiting resistor. LED- is the cathode of the back-light LED and this pin must be connected to ground.

3.5 16×2 LCD module commands

16×2 LCD module has a set of preset command instructions. Each command will make the module to do a particular task. The commonly used commands and their function are given in the table 3.5.1 below.

Table:3Functions of LCD

Command	Function
0F	LCD ON, Cursor ON, Cursor blinking ON
01	Clear screen
02	Return home
04	Decrement cursor
06	Increment cursor
0E	Display ON, Cursor blinking OFF
80	Force cursor to the beginning of 1 st line
C0	Force cursor to the beginning of 2 nd line
38	Use 2 lines and 5×7 matrix
83	Cursor line 1 position 3
3C	Activate second line
08	Display OFF, Cursor OFF
C1	Jump to second line, position1
OC	Display ON, Cursor OFF
C1	Jump to second line, position1
C2	Jump to second line, position2

3.5.1 LCD Initialization.

The steps that has to be done for initializing the LCD display is given below and these steps are common for almost all applications.

- Send 38H to the 8-bit data line for initialization
- Send 0FH for making LCD ON, cursor ON and cursor blinking ON.
- Send 06H for incrementing cursor position.
- Send 01H for clearing the display and return the cursor.

Sending data to the LCD

The steps for sending data to the LCD module is given below. I have already said that the LCD module has pins namely RS, R/W and E. It is the logic state of these pins that make the module to determine whether a given data input is a command or data to be displayed.

- Make R/W low.
- Make RS=0 if data byte is a command and make RS=1 if the data byte is a data to be displayed.
- Place data byte on the data register.
- Pulse E from high to low.

3.6 Relay

A relay is an electrically operated switch. Many relays use an electromagnet to mechanically Operate a switch, but other operating principles are also used, such as solid-state relays. Relays are used where it is necessary to control a circuit by a low-power signal (with complete electrical isolation between control and controlled circuits), or where several circuits must be controlled by one signal. The first relays were used in long distance telegraph Repeat above steps for sending another data circuits as amplifiers: they repeated the signal coming in from one circuit and re-transmitted it on another circuit. Relays were used extensively in telephone exchanges and early computers to perform logical operations.

A type of relay that can handle the high power required to directly control an electric motor or other loads is called a contactor. Solid-state relays control power circuits with no

moving parts, instead using a semiconductor device to perform switching. Relays with calibrated operating characteristics and sometimes multiple operating coils are used to protect electrical circuits from overload or faults; in modern electric power systems these functions are performed by digital instruments still called "protective relays".



Fig.3.6 Electromagnetic relay

3.6.1 Basic design and Operation

A simple electromagnetic relay consists of a coil of wire wrapped around a soft iron core, an iron yoke which provides a low reluctance path for magnetic flux, a movable iron armature, and one or more sets of contacts (there are two in the relay pictured). The armature is hinged to the yoke and mechanically linked to one or more sets of moving contacts. It is held in place by a spring so that when the relay is de-energized there is an air gap in the magnetic circuit. In this condition, one of the two sets of contacts in the relay pictured is closed, and the other set is open. Other relays may have more or fewer sets of contacts depending on their function. The relay in the picture also has a wire connecting the armature to the yoke. This ensures continuity of the circuit between the moving contacts on the armature, and the circuit track on the printed circuit board (PCB) via the yoke, which is soldered to the PCB.

When an electric current is passed through the coil it generates a magnetic field that activates the armature, and the consequent movement of the movable contact(s) either makes or breaks (depending upon construction) a connection with a fixed contact. If the set of contacts was closed when the relay was de-energized, then the movement opens the contacts and breaks the connection, and vice versa if the contacts were open. When the current to the coil is switched off, the armature is returned by a force, approximately half as strong as the

magnetic force, to its relaxed position. Usually this force is provided by a spring, but gravity is also used commonly in industrial motor starters. Most relays are manufactured to operate quickly. In a low-voltage application this reduces noise; in a high voltage or current application it reduces arcing.

When the coil is energized with direct current, a diode is often placed across the coil to dissipate the energy from the collapsing magnetic field at deactivation, which would otherwise generate a voltage spike dangerous to semiconductor circuit components. Such diodes were not widely used before the application of transistors as relay drivers, but soon became ubiquitous as early germanium transistors were easily destroyed by this surge. Some automotive relays include a diode inside the relay case.

If the relay is driving a large, or especially a reactive load, there may be a similar problem of surge currents around the relay output contacts. In this case a snubber circuit (a capacitor and resistor in series) across the contacts may absorb the surge. Suitably rated capacitors and the associated resistor are sold as a single packaged component for this commonplace use.

If the coil is designed to be energized with alternating current (AC), some method is used to split the flux into two out-of-phase components which add together, increasing the minimum pull on the armature during the AC cycle. Typically, this is done with a small copper "shading ring" crimped around a portion of the core that creates the delayed, out-of-phase component, which holds the contacts during the zero crossings of the control voltage.

3.6.2 Pole and Throw

Relays have the exact working of a switch. So, the same concept is also applied. A relay is said to switch one or more poles. Each pole has contacts that can be thrown in mainly three ways. They are

- **Normally Open Contact (NO)** – NO contact is also called a make contact. It closes the circuit when the relay is activated. It disconnects the circuit when the relay is inactive.
- **Normally Closed Contact (NC)** – NC contact is also known as break contact. This is opposite to the NO contact. When the relay is activated, the circuit disconnects. When the relay is deactivated, the circuit connects.

- **Change-over (CO) / Double-throw (DT) Contacts** – This type of contacts are used to control two types of circuits. They are used to control a NO contact and also a NC contact with a common terminal. According to their type they are called by the names break before make and make before break contacts

Relays are also named with designations like

- **Single Pole Single Throw (SPST)** – This type of relay has a total of four terminals. Out of these two terminals can be connected or disconnected. The other two terminals are needed for the coil.
- **Single Pole Double Throw (SPDT)** – This type of a relay has a total of five terminals. Out of these two are the coil terminals. A common terminal is also included which connects to either of two others.
- **Double Pole Single Throw (DPST)** – This relay has a total of six terminals. These terminals are further divided into two pairs. Thus they can act as two SPST's which are actuated by a single coil. Out of the six terminals two of them are coil terminals.
- **Double Pole Double Throw (DPDT)** – This is the biggest of all. It has mainly eight relay terminals. Out of these two rows are designed to be change over terminals. They are designed to act as two SPDT relays which are actuated by a single coil.

3.6.3 Relay Applications

- Relays are used to realize logic functions. They play a very important role in providing safety critical logic.
- Relays are used to provide time delay functions. They are used to time the delay open and delay close of contacts.
- Relays are used to control high voltage circuits with the help of low voltage signals. Similarly, they are used to control high current circuits with the help of low current signals.

3.7. Induction Motor

3.7.1 Principles of operation

In both induction and synchronous motors, the AC power supplied to the motor's stator creates a magnetic field that rotates in time with the AC oscillations. Whereas a

synchronous motor's rotor turns at the same rate as the stator field, an induction motor's rotor rotates at a slower speed than the stator field. The induction motor stator's magnetic field is therefore changing or rotating relative to the rotor. This induces an opposing current in the induction motor's rotor, in effect the motor's secondary winding, when the latter is short-circuited or closed through an external impedance. The rotating magnetic flux induces currents in the windings of the rotor; in a manner similar to currents induced in a transformer's secondary winding(s). The currents in the rotor windings in turn create magnetic fields in the rotor that react against the stator field. Due to Lenz's Law, the direction of the magnetic field created will be such as to oppose the change in current through the rotor windings. The cause of induced current in the rotor windings is the rotating stator magnetic field, so to oppose the change in rotor-winding currents the rotor will start to rotate in the direction of the rotating stator magnetic field. The rotor accelerates until the magnitude of induced rotor current and torque balances the applied load. Since rotation at synchronous speed would result in no induced rotor current, an induction motor always operates slower than synchronous speed. The difference, or "slip," between actual and synchronous speed varies from about 0.5 to 5.0% for standard Design B torque curve induction motors.^[24] The induction machine's essential character is that it is created solely by induction instead of being separately excited as in synchronous or DC machines or being self-magnetized as in permanent magnet motors.

For rotor currents to be induced, the speed of the physical rotor must be lower than that of the stator's rotating magnetic field (n_s); otherwise the magnetic field would not be moving relative to the rotor conductors and no currents would be induced. As the speed of the rotor drops below synchronous speed, the rotation rate of the magnetic field in the rotor increases, inducing more current in the windings and creating more torque. The ratio between the rotation rate of the magnetic field induced in the rotor and the rotation rate of the stator's rotating field is called slip. Under load, the speed drops and the slip increase enough to create sufficient torque to turn the load. For this reason, induction motors are sometimes referred to as asynchronous motors. An induction motor can be used as an induction generator, or it can be unrolled to form a linear induction motor which can directly generate linear motion.

3.8 Transformer

A transformer is an electrical device that transfers electrical energy between two or more circuits through electromagnetic induction. Electromagnetic induction produces an electromotive force across a conductor which is exposed to time varying magnetic fields. Commonly, transformers are used to increase or decrease the voltages of alternating current in electric power applications. A varying current in the transformer's primary winding creates a varying magnetic flux in the transformer core and a varying magnetic field impinging on the transformer's secondary winding. This varying magnetic field at the secondary winding induces a varying electromotive force (EMF) or voltage in the secondary winding due to electromagnetic induction.

A Transformer takes in electricity at a higher voltage and lets it run through lots of coils wound around an iron core. “A single-phase Transformer can operate to either increase or decrease the voltage applied to the primary winding. Because the current is alternating, the magnetism in the core is also alternating. Also, around the core is an output wire with fewer coils. The magnetism changing back and forth makes a current in the wire. Having fewer coils means less voltage. When it is used to “decrease” the voltage on the secondary winding with respect to the primary it is called a Step-down Transformer. When a Transformer is used to “increase” the voltage on its secondary winding with respect to the primary, it is called a Step-up Transformer. However, a third condition exists in which a transformer produces the same voltage on its secondary as is applied to its primary winding. In other words, its output is identical with respect to input. This type of Transformer is called an “Impedance Transformer” and is mainly used for impedance matching or the isolation of adjoining electrical circuits.



Fig.3.7 Step-down transformer

3.8.1 Types of transformer

- Various specific electrical application designs require a variety of transformer types. Although they all share the basic characteristic transformer principles, they are customized in construction or electrical properties for certain installation requirements or circuit conditions.
- Autotransformer: Transformer in which part of the winding is common to both primary and secondary circuits.
- Capacitor voltage transformer: Transformer in which capacitor divider is used to reduce high voltage before application to the primary winding.
- Distribution transformer, power transformer: International standards make a distinction in terms of distribution transformers being used to distribute energy from transmission lines and networks for local consumption and power transformers being used to transfer electric energy between the generator and distribution primary circuits.
- Phase angle regulating transformer: A specialised transformer used to control the flow of real power on three-phase electricity transmission networks.
- Scott-T transformer: Transformer used for phase transformation from three-phase to two-phase and vice versa.

3.8.2 Applications of transformer

Transformers are used to increase (or step-up) voltage before transmitting electrical energy over long distances through wires. Wires have resistance which loses energy through joule heating at a rate corresponding to square of the current. By transforming power to a higher voltage transformer enable economical transmission of power and distribution. Consequently, transformers have shaped the electricity supply industry, permitting generation to be located remotely from points of demand. All but a tiny fraction of the world's electrical power has passed through a series of transformers by the time it reaches the consumer.

Transformers are also used extensively in electronic products to decrease (or step-down) the supply voltage to a level suitable for the low voltage circuits they contain. The transformer also electrically isolates the end user from contact with the supply voltage.

Signal and audio transformers are used to couple stages of amplifiers and to match devices such as microphones and record players to the input of amplifiers. Audio transformers allowed telephone circuits to carry on a two-way conversation over a single pair of wires. A balun transformer converts a signal that is referenced to ground to a signal that has balanced voltages to ground, such as between external cables and internal circuits.

3.9 Voltage Regulators

A voltage regulator is designed to automatically maintain a constant voltage level. A voltage regulator may be a simple "feed-forward" design or may include negative feedback control loops. It may use an electromechanical mechanism, or electronic components. Depending on the design, it may be used to regulate one or more AC or DC voltages.

Electronic voltage regulators are found in devices such as computer power supplies where they stabilize the DC voltages used by the processor and other elements. In automobile alternators and central power station generator plants, voltage regulators control the output of the plant. In an electric power distribution system, voltage regulators may be installed at a substation or along distribution lines so that all customers receive steady voltage independent of how much power is drawn from the line.

The 78xx (sometimes L78xx, LM78xx, MC78xx...) is a family of self-contained fixed linear voltage regulator integrated circuits. The 78xx family is commonly used in electronic circuits requiring a regulated power supply due to their ease-of-use and low cost. For ICs within the family, the xx is replaced with two digits, indicating the output voltage (for example, the 7805 has a 5-volt output, while the 7812 produces 12 volts). The 78xx line are positive voltage regulators: they produce a voltage that is positive relative to a common ground. There is a related line of 79xx devices which are complementary negative voltage regulators. 78xx and 79xx ICs can be used in combination to provide positive and negative supply voltages in the same circuit.

78xx ICs have three terminals and are commonly found in the TO-220 form factor, although they are available in surface-mount, TO-92, and TO-3 packages. These devices support an input voltage anywhere from around 2.5 volts over the intended output voltage up to a maximum of 35 to 40 volts depending on the model, and typically provide 1 or 1.5

amperes of current (though smaller or larger packages may have a lower or higher current rating).



Fig.3.8 Voltage Regulators

3.9.1 Advantages

- 78xx series ICs do not require additional components to provide a constant, regulated source of power, making them easy to use, as well as economical and efficient uses of space. Other voltage regulators may require additional components to set the output voltage level, or to assist in the regulation process. Some other designs (such as a switched-mode power supply) may need substantial engineering expertise to implement.
- 78xx series ICs have built-in protection against a circuit drawing too much current. They have protection against overheating and short-circuits, making them quite robust in most applications. In some cases, the current-limiting features of the 78xx devices can provide protection not only for the 78xx itself, but also for other parts of the circuit.

7805 is a voltage regulator integrated circuit. It is a member of 78xx series of fixed linear voltage regulator ICs. The voltage source in a circuit may have fluctuations and would not give the fixed voltage output. The voltage regulator IC maintains the output voltage at a constant value. The xx in 78xx indicates the fixed output voltage it is designed to provide. 7805 provides +5V regulated power supply. Capacitors of suitable values can be connected at input and output pins depending upon the respective voltage levels.

Pin Description:**Table.4** Voltage regulator functions

Pin No	Function	Name
1	Input voltage (5V-18V)	Input
2	Ground (0V)	Ground
3	Regulated output; 5V (4.8V-5.2V)	Output

3.10 Transistors

A transistor is a semiconductor device used to amplify or switch electronic signals and electrical power. It is composed of semiconductor material with at least three terminals for connection to an external circuit. A voltage or current applied to one pair of the transistor's terminals changes the current through another pair of terminals. Because the controlled (output) power can be higher than the controlling (input) power, a transistor can amplify a signal. Today, some transistors are packaged individually, but many more are found embedded in integrated circuits.

3.10.1 Symbols, Pins, and Construction

Transistors are fundamentally three-terminal devices. On a bi-polar junction transistor (BJT), those pins are labeled collector (C), base (B), and emitter (E).

The only difference between an NPN and PNP is the direction of the arrow on the emitter. The arrow on an NPN points out, and on the PNP it points in.

3.10.2 Transistor Construction

Transistors rely on semiconductors to work their magic. A semiconductor is a material that's not quite a pure conductor (like copper wire) but also not an insulator (like air). The conductivity of a semiconductor – how easily it allows electrons to flow – depends on variables like temperature or the presence of more or less electrons. Let's look briefly under the hood of a transistor. Don't worry we won't dig too deeply into quantum physics.



Fig.3.9 Transistors

3.10.3 Transistor as Two Diodes

Transistors are kind of like an extension of another semiconductor component: diodes. In way transistors are just two diodes with their cathodes (or anodes) tied together:

The diode connecting base to emitter is the important one here; it matches the direction of the arrow on the schematic symbol, and shows you which way current is intended to flow through the transistor.

The diode representation is a good place to start, but it's far from accurate. Don't base your understanding of a transistor's operation on that model (and definitely don't try to replicate it on a breadboard, it won't work). There's a whole lot of weird quantum physics level stuff controlling the interactions between the three terminals.

Then Bipolar Transistors have the ability to operate within three different regions:

- Active Region – the transistor operates as an amplifier and $I_c = \beta \cdot I_b$
- Saturation – the transistor is “Fully-ON” operating as a switch and $I_c = I(\text{saturation})$
- Cut-off – the transistor is “Fully-OFF” operating as a switch and $I_c = 0$

The Bipolar Transistor basic construction consists of two PN-junctions producing three connecting terminals with each terminal being given a name to identify it from the other two. These three terminals are known and labeled as the Emitter (E), the Base (B) and the Collector (C) respectively.

Bipolar Transistors are current regulating devices that control the amount of current flowing through them in proportion to the amount of biasing voltage applied to their base terminal acting like a current-controlled switch. The principle of operation of the two transistor types PNP and NPN, is exactly the same the only difference being in their biasing and the polarity of the power supply for each type.

3.10.4 Basic NPN Transistor Switching Circuit

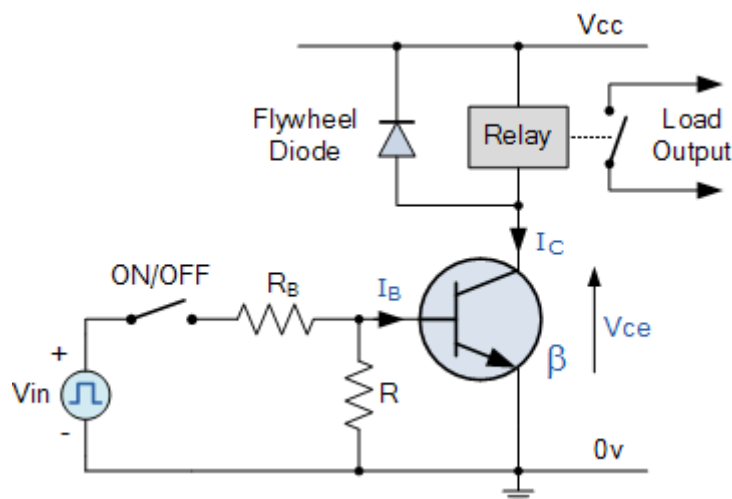


Fig.3.10 NPN Transistor Switching Circuit

The circuit resembles that of the Common Emitter circuit we looked at in the previous tutorials. The difference this time is that to operate the transistor as a switch the transistor needs to be turned either fully “OFF” (cut-off) or fully “ON” (saturated). An ideal transistor switch would have infinite circuit resistance between the Collector and Emitter when turned “fully-OFF” resulting in zero current flowing through it and zero resistance between the Collector and Emitter when turned “fully-ON”, resulting in maximum current flow.

In practice when the transistor is turned “OFF”, small leakage currents flow through the transistor and when fully “ON” the device has a low resistance value causing a small saturation voltage (V_{CE}) across it. Even though the transistor is not a perfect switch, in both the cut-off and saturation regions the power dissipated by the transistor is at its minimum.

In order for the Base current to flow, the Base input terminal must be made more positive than the Emitter by increasing it above the 0.7 volts needed for a silicon device. By varying this Base-Emitter voltage V_{BE} , the Base current is also altered and which in turn

controls the amount of Collector current flowing through the transistor as previously discussed.

When maximum Collector current flows the transistor is said to be Saturated. The value of the Base resistor determines how much input voltage is required and corresponding Base current to switch the transistor fully “ON”.

Then to summaries when using a Transistor as a Switch the following conditions apply:

- Transistor switches can be used to switch and control lamps, relays or even motors.
- When using the bipolar transistor as a switch they must be either “fully-OFF” or “fully-ON”.
- Transistors that are fully “ON” are said to be in their Saturation region.
- Transistors that are fully “OFF” are said to be in their Cut-off region.
- When using the transistor as a switch, a small Base current controls a much larger Collector load current.
- When using transistors to switch inductive loads such as relays and solenoids, a “Flywheel Diode” is used.
- When large currents or voltages need to be controlled, Darlington Transistors can be used.

3.11 Resistor

A resistor is a passive two-terminal electrical component that implements electrical resistance as a circuit element. Resistors act to reduce current flow, and, at the same time, act to lower voltage levels within circuits. In electronic circuits, resistors are used to limit current flow, to adjust signal levels, bias active elements, and terminate transmission lines among other uses. High-power resistors, that can dissipate many watts of electrical power as heat, may be used as part of motor controls, in power distribution systems, or as test loads for generators. Fixed resistors have resistances that only change slightly with temperature, time or operating voltage. Variable resistors can be used to adjust circuit elements (such as a volume control or a lamp dimmer), or as sensing devices for heat, light, humidity, force, or chemical activity.

Resistors are common elements of electrical networks and electronic circuits and are ubiquitous in electronic equipment. Practical resistors as discrete components can be composed of various compounds and forms. Resistors are also implemented within integrated circuits.

Resistor units

The electrical resistance of a resistor is measured in ohms. The symbol for an ohm is the greek capital-omega: Ω . The (somewhat roundabout) definition of 1Ω is the resistance between two points where 1 volt (1V) of applied potential energy will push 1 ampere (1A) of current.

As SI units go, larger or smaller values of ohms can be matched with a prefix like kilo-, mega-, or giga-, to make large values easier to read. It's very common to see resistors in the kilohm ($k\Omega$) and megaohm ($M\Omega$) range (much less common to see miliohm ($m\Omega$) resistors). For example, a $4,700\Omega$ resistor is equivalent to a $4.7k\Omega$ resistor, and a $5,600,000\Omega$ resistor can be written as $5,600k\Omega$ or (more commonly as) $5.6M\Omega$.



Fig.3.11 Variable resistor

3.11.1 Types of Resistors

Resistors come in a variety of shapes and sizes. They might be through-hole or surface-mount. They might be a standard, static resistor, a pack of resistors, or a special variable resistor.

3.11.2 Termination and mounting

Resistors will come in one of two termination-types: through-hole or surface-mount. These types of resistors are usually abbreviated as either PTH (plated through-hole) or SMD/SMT (surface-mount technology or device).

Through-hole resistors come with long, pliable leads which can be stuck into a breadboard or hand-soldered into a prototyping board or printed circuit board (PCB). These resistors are usually more useful in breadboarding, prototyping, or in any case where you'd rather not solder tiny, little 0.6mm-long SMD resistors. The long leads usually require trimming, and these resistors are bound to take up much more space than their surface-mount counterparts.

The most common through-hole resistors come in an axial package. The size of an axial resistor is relative to its power rating. A common $\frac{1}{2}W$ resistor measures about 9.2mm across, while a smaller $\frac{1}{4}W$ resistor is about 6.3mm long.

Surface-mount resistors are usually tiny black rectangles, terminated on either side with even smaller, shiny, silver, conductive edges. These resistors are intended to sit on top of PCBs, where they're soldered onto mating landing pads. Because these resistors are so small, they're usually set into place by a robot, and sent through an oven where solder melts and holds them in place.

SMD resistors come in standardized sizes; usually either 0805 (0.8mm long by 0.5mm wide), 0603, or 0402. They're great for mass circuit-board-production, or in designs where space is a precious commodity. They take a steady, precise hand to manually solder, though!

3.11.3 Measurement

The value of a resistor can be measured with an ohmmeter, which may be one function of a multimeter. Usually, probes on the ends of test leads connect to the resistor. A simple ohmmeter may apply a voltage from a battery across the unknown resistor (with an internal resistor of a known value in series) producing a current which drives a meter movement. The current, in accordance with Ohm's law, is inversely proportional to the sum of the internal resistance and the resistor being tested, resulting in an analog meter scale which is very non-linear, calibrated from infinity to 0 ohms. A digital multimeter, using active electronics, may

instead pass a specified current through the test resistance. The voltage generated across the test resistance in that case is linearly proportional to its resistance, which is measured and displayed. In either case the low-resistance ranges of the meter pass much more current through the test leads than do high-resistance ranges, in order for the voltages present to be at reasonable levels (generally below 10 volts) but still measurable.

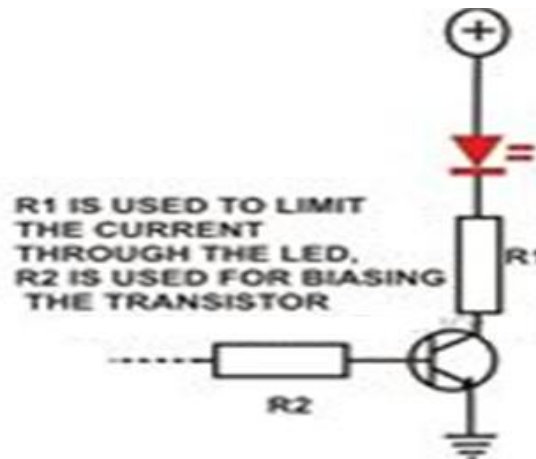


Fig.3.12 Current through LED

3.11.4 Applications of Resistors

In electronic circuits, resistors play an important role to limit the current and provide only the required biasing to the vital active parts like the transistors and the ICs. We will try to find out what is the function of a resistor in electronics through the following illustrations:

Transistor Biasing: Through one of my previous articles you must have acquired a good knowledge regarding transistors. A transistor basically needs a small base voltage (>0.6) to make a large voltage flow through its collector/ emitter terminals. But the base of a transistor is quite vulnerable to high currents, so a resistor is incorporated here to limit the current and provide a safe biasing voltage.

The value of the base resistor of a transistor may be calculated through the below given formula:

$$R = (V - 0.6) \cdot H_{fe} / I,$$

Here V = source voltage to the base resistor, I = the collector load current, H_{fe} = forward gain of a transistor (150 nominal) and 0.6 = minimum transistor biasing voltage.

LED Current Limit:

Just like transistors, LEDs too are very sensitive to high currents. A resistor when placed in series with the LEDs regulates a proper flow of current through them. To calculate the value of a series LED resistor, the following formula may be used:

$$R = (V - (N \cdot V_{LED})) / I$$

3.12 Diodes

A diode is an electronic component with two electrodes (connectors). It allows electricity to go through it only in one direction.

Diodes can be used to convert alternating current to direct current (Diode bridge). They are often used in power supplies and sometimes to decode amplitude modulation radio signals (like in a crystal radio). Light-emitting diodes (LEDs) are a type of diode that produce light. Today, the most common diodes are made from semiconductor materials such as silicon or sometimes germanium.

3.12.1 Construction

Semiconductor diodes are made of two types of semiconductors connected to each other. One type has atoms with extra electrons (called the n-side). The other type has atoms that want electrons (called the p-side). Because of this, the electricity will flow easily from the side with too many electrons to the side with too few. However, electricity will not flow easily in the reverse direction. Silicon with arsenic dissolved in it makes a good n-side semiconductor, while silicon with aluminum dissolved in it makes a good p-side semiconductor, but other materials can also work. The connector to the n-side is called the cathode, the connector to the p-side is called the anode.

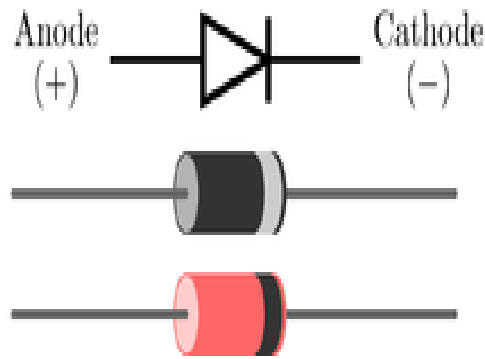


Fig.3.13Semiconductor diodes

3.12.2 Function of a diode

Positive voltage at p-side

If you give positive voltage to the p-side and negative voltage to the n-side, the electrons in the n-side will want to go to the positive voltage at the p-side and the holes of the p-side will want to go to the negative voltage at the n-side. Because of this, current flow is able to exist, but it takes a certain amount of voltage to get this started (very small amount of voltage is not enough to get the electric current to flow). This is called the cut-in voltage. The cut-in voltage of a silicon diode is at about 0.7 V. A germanium diode needs a cut-in voltage at about 0.3 V.

Negative voltage at p-side

If you instead give negative voltage to the p-side and positive voltage to the n-side, the electrons of the n-side want to go to the positive voltage source instead of the other side of the diode. Same thing happens on the p-side. So, current will not flow between the two sides of the diode. Increasing the voltage will eventually force electric current to flow (this is the break-down voltage). Many diodes will be destroyed by a reverse flow but some are made that can survive it.

3.12.3 Types of diodes

Here are some common semiconductor diode symbols used in schematic diagrams:

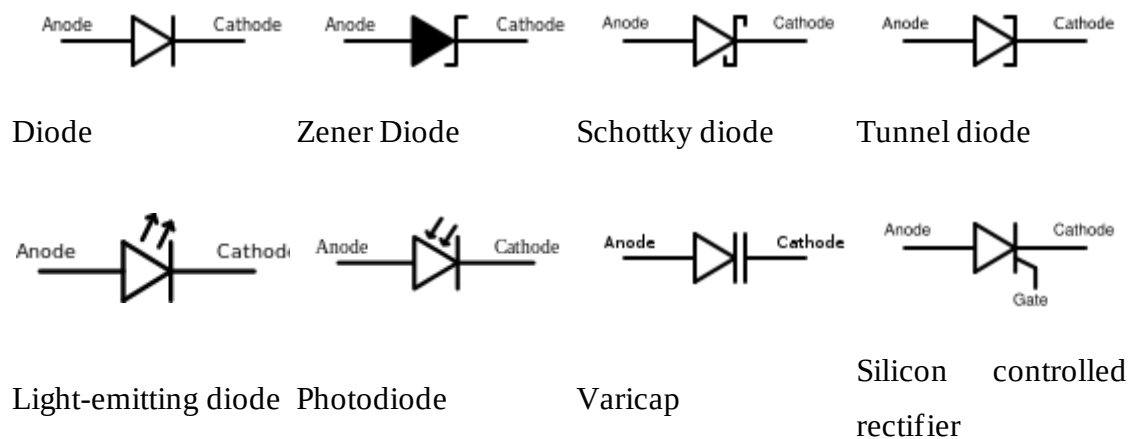


Fig.3.14 Types of diodes

3.12.4 Functions of Diodes

The most common function of a diode is to allow an electric current to pass in one direction (called the diode's forward direction), while blocking current in the opposite direction (the *reverse* direction). Thus, the diode can be viewed as an electronic version of a check valve. This unidirectional behavior is called rectification, and is used to convert alternating current to direct current, including extraction of modulation from radio signals in radio receivers—these diodes are forms of rectifiers.

However, diodes can have more complicated behavior than this simple on–off action, due to their nonlinear current-voltage characteristics. Semiconductor diodes begin conducting electricity only if a certain threshold voltage or cut-in voltage is present in the forward direction (a state in which the diode is said to be *forward-biased*). The voltage drop across a forward-biased diode varies only a little with the current, and is a function of temperature; this effect can be used as a temperature sensor or as a voltage reference.

A semiconductor diode's current–voltage characteristic can be tailored by selecting the semiconductor materials and the doping impurities introduced into the materials during manufacture. These techniques are used to create special-purpose diodes that perform many different functions. For example, diodes are used to regulate voltage (Zener diodes), to

protect circuits from high voltage surges (avalanche diodes), to electronically tune radio and TV receivers (varactor diodes), to generate radio-frequency oscillations (tunnel diodes, Gunn diodes, IMPATT diodes), and to produce light (light-emitting diodes). Tunnel, Gunn and IMPATT diodes exhibit negative resistance, which is useful in microwave and switching circuits.

3.12.5 Applications of Diodes

- Radio demodulation
- Power conversion
- Over-voltage protection
- Logic gates
- Temperature measurements

3.13 TSOP 1738 – IR Receiver

The TSOP 1738 is a member of IR remote control receiver series. This IR sensor module consists of a PIN diode and a pre amplifier which are embedded into a single package. The output of TSOP is active low and it gives +5V in off state. When IR waves, from a source, with a centre frequency of 38 kHz incident on it, its output goes low.

Lights coming from sunlight, fluorescent lamps etc. may cause disturbance to it and result in undesirable output even when the source is not transmitting IR signals. A bandpass filter, an integrator stage and an automatic gain control are used to suppress such disturbances.

TSOP module has an inbuilt control circuit for amplifying the coded pulses from the IR transmitter. A signal is generated when PIN photodiode receives the signals. This input signal is received by an automatic gain control (AGC). For a range of inputs, the output is fed back to AGC in order to adjust the gain to a suitable level. The signal from AGC is passed to a band pass filter to filter undesired frequencies. After this, the signal goes to a demodulator and this demodulated output drives an npn transistor. The collector output of the transistor is obtained at pin 3 of TSOP module.

Members of TSOP17xx series are sensitive to different centre frequencies of the IR spectrum. For example, TSOP1738 is sensitive to 38 kHz whereas TSOP1740 to 40 kHz centre frequency.

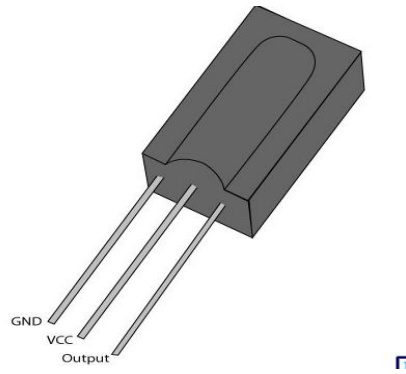


Fig.3.15 TSOP 1738 – IR Receiver

3.14 Optocoupler

An Optocoupler, also known as an Opto-isolator or Photo-coupler, is an electronic component that interconnects two separate electrical circuits by means of a light sensitive optical interface.

The basic design of an Optocoupler consists of an LED that produces infra-red light and a semiconductor photo-sensitive device that is used to detect the emitted infra-red beam. Both the LED and photo-sensitive device are enclosed in a light-tight body or package with metal legs for the electrical connections as shown.

An Optocoupler or opto-isolator consists of a light emitter, the LED and a light sensitive receiver which can be a single photo-diode, photo-transistor, photo-resistor, photo-SCR, or a photo-TRIAC and the basic operation of an optocoupler is very simple to understand.

Phototransistor Optocoupler

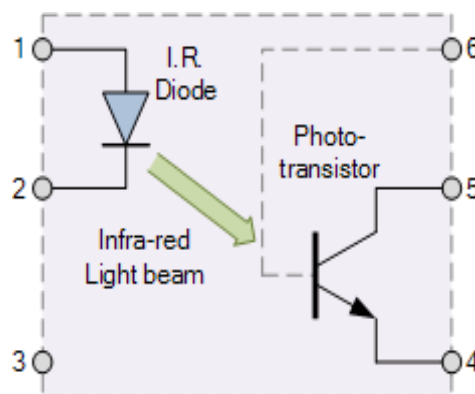


Fig. 3.16 Optocoupler

Assume a photo-transistor device as shown. Current from the source signal passes through the input LED which emits an infra-red light whose intensity is proportional to the electrical signal. This emitted light falls upon the base of the photo-transistor, causing it to switch-ON and conduct in a similar way to a normal bipolar transistor.

The base connection of the photo-transistor can be left open for maximum sensitivity or connected to ground via a suitable external resistor to control the switching sensitivity making it more stable.

When the current flowing through the LED is interrupted, the infra-red emitted light is cut-off, causing the photo-transistor to cease conducting. The photo-transistor can be used to switch current in the output circuit. The spectral response of the LED and the photo-sensitive device are closely matched being separated by a transparent medium such as glass, plastic or air. Since there is no direct electrical connection between the input and output of an optocoupler, electrical isolation up to 10kV is achieved.

Optocouplers are available in four general types, each one having an infra-red LED source but with different photo-sensitive devices. The four optocouplers are called the: Photo-transistor, Photo-darlington, Photo-SCR and Photo-triac as shown below.

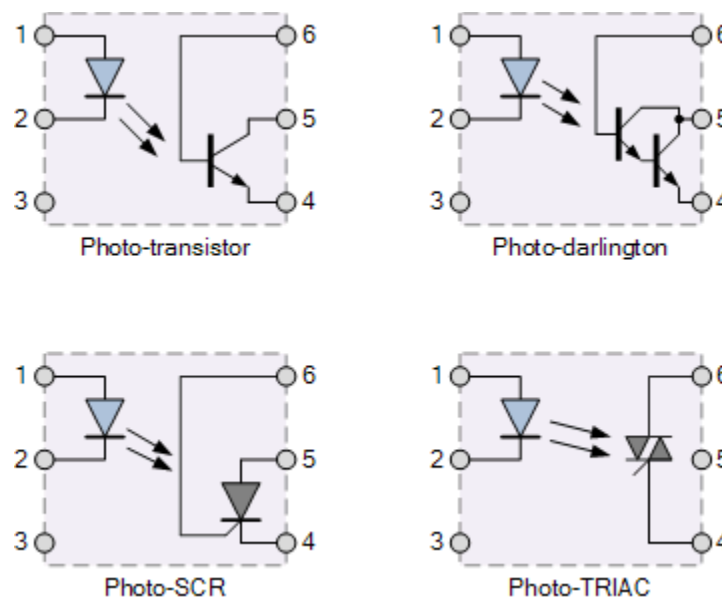


Fig.3.17 Types of Optocoupler

The photo-transistor and photo-darlington devices are mainly for use in DC circuits while the photo-SCR and photo-triac allow AC powered circuits to be

controlled. There are many other kinds of source-sensor combinations, such as LED-photodiode, LED-LASER, lamp-photoresistor pairs, reflective and slotted optocouplers.

Simple homemade optocouplers can be constructed by using individual components. An Led and a photo-transistor are inserted into a rigid plastic tube or encased in heat-shrinkable tubing as shown. The advantage of this home-made optocoupler is that tubing can be cut to any length you want and even bent around corners. Obviously, tubing with a reflective inner would be more efficient than dark black tubing.

Home-made Optocoupler

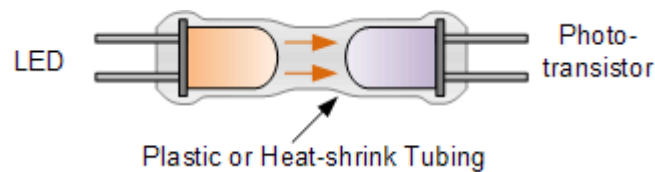


Fig.3.18 Photo transistor

3.14.1 Optocoupler Applications

Optocouplers and opto-isolators can be used on their own, or to switch a range of other larger electronic devices such as transistors and triacs providing the required electrical isolation between a lower voltage control signal and the higher voltage or current output signal. Common applications for optocouplers include microprocessor input/output switching, DC and AC power control, PC communications, signal isolation and power supply regulation which suffer from current ground loops, etc. The electrical signal being transmitted can be either analogue (linear) or digital (pulses).

In this application, the optocoupler is used to detect the operation of the switch or another type of digital input signal. This is useful if the switch or signal being detected is within an electrically noisy environment. The output can be used to operate an external circuit, light or as an input to a PC or microprocessor.

An Optotransistor DC Switch

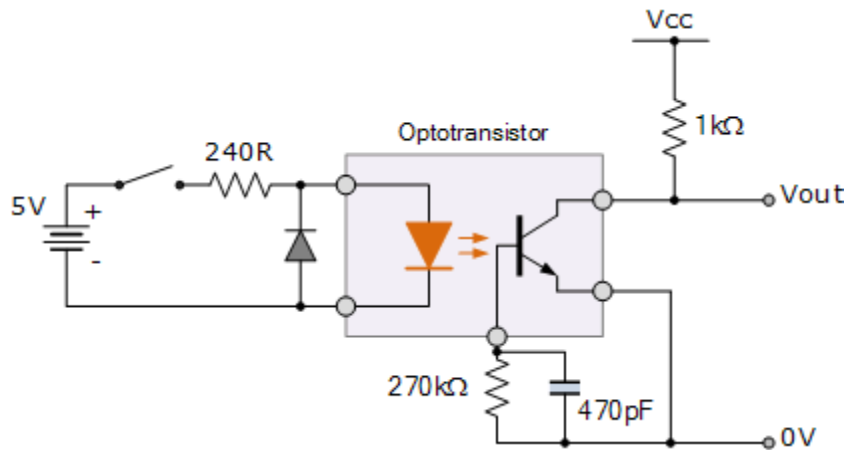


Fig.3.19 larger triac via a current limiting resistor

As well as detecting DC signals and data, Opto-triac isolators are also available which allow AC powered equipment and mains lamps to be controlled. Opto-coupled triacs such as the MOC 3020, have voltage ratings of about 400 volts making them ideal for direct mains connection and a maximum current of about 100mA. For higher powered loads, the opto-triac may be used to provide the gate pulse to another larger triac via a current limiting resistor as shown.

Triac Optocoupler Application

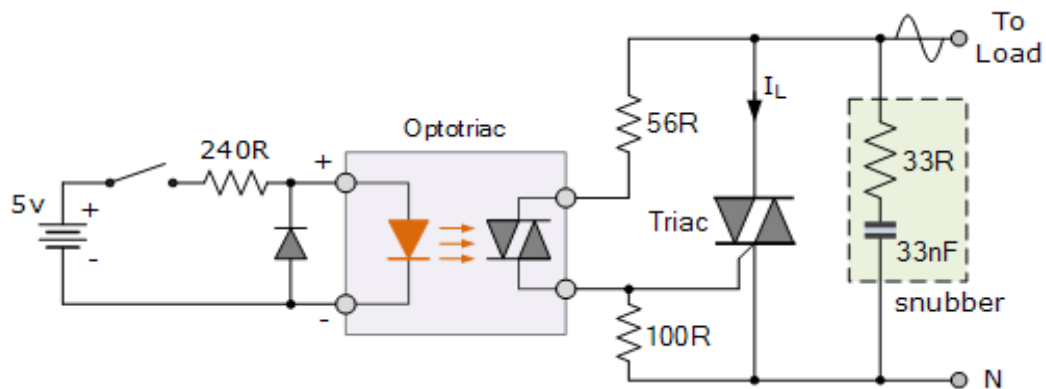


Fig.3.20Triac Optocoupler

This type of optocoupler configuration forms the basis of a very simple solid state relay application which can be used to control any AC mains powered load such as lamps and motors. Also unlike a thyristor (SCR), a triac is capable of conducting in both halves of the mains AC cycle with zero-crossing detection allowing the load to receive full power without the heavy inrush currents when switching inductive loads.

Optocouplers and Opto-isolators are great electronic devices that allow devices such as power transistors and triacs to be controlled from a PC's output port, digital switch or from a low voltage data signal such as that from a logic gate. The main advantage of opto-couplers is their high electrical isolation between the input and output terminals allowing relatively small digital signals to control much large AC voltages, currents and power.

An optocoupler can be used with both DC and AC signals with optocouplers utilizing a SCR (thyristor) or triac as the photo-detecting device are primarily designed for AC power-control applications. The main advantage of photo-SCRs and photo-triacs is the complete isolation from any noise or voltage spikes present on the AC power supply line as well as zero-crossing detection of the sinusoidal waveform which reduces switching and inrush currents protecting any power semiconductors used from thermal stress and shock.

3.15 Triac

TRIAC, from triode for alternating current, is a generic trademark for a three terminal electronic component that conducts current in either direction when triggered. Its formal name is bidirectional triode thyristor or bilateral triode thyristor. A thyristor is analogous to a relay in that a small voltage and current can control a much larger voltage and current. The illustration on the right shows the circuit symbol for a TRIAC where A1 is Anode 1, A2 is Anode 2, and G is Gate. Anode 1 and Anode 2 are normally termed Main Terminal 1 (MT1) and Main Terminal 2 (MT2) respectively.

TRIACs are a subset of thyristors and are related to silicon controlled rectifiers (SCRs). However, unlike SCRs, which are unidirectional devices and only conduct current in one direction, TRIACs are bidirectional and conduct current in both directions. Another difference is that SCRs can only be triggered by a positive current at their gate, but, in general, TRIACs can be triggered by either a positive or negative current at their gate, although some special types cannot be triggered by one of the combinations. To create a triggering current for an SCR a positive voltage has to be applied to the gate but for a TRIAC either a positive or negative voltage can be applied to the gate. In all three cases the voltage and current are with respect to MT1.

Once triggered, SCRs and thyristors continue to conduct, even if the gate current ceases, until the main current drops below a certain level called the holding current.

Gate turn-off thyristors (GTOs) are similar to TRIACs but provide more control by turning off when the gate signal ceases. TRIACs bidirectionality makes them convenient switches for alternating-current (AC). In addition, applying a trigger at a controlled phase angle of the AC in the main circuit allows control of the average current flowing into a load (phase control). This is commonly used for controlling the speed of induction motors, dimming lamps, and controlling electric heaters.

3.15.1 Operation

To understand how TRIACs work, consider the triggering in each of the four quadrants. The four quadrants are illustrated in Figure 1, and depend on the gate and MT2 voltages with respect to MT1. Main Terminal 1 (MT1) and Main Terminal 2 (MT2) are also referred to as Anode 1 (A1) and Anode 2 (A2) respectively.^[1]

The relative sensitivity depends on the physical structure of a particular triac, but as a rule, quadrant I is the most sensitive (least gate current required), and quadrant 4 is the least sensitive (most gate current required).

In quadrants 1 and 2, MT2 is positive, and current flows from MT2 to MT1 through P, N, P and N layers. The N region attached to MT2 does not participate significantly. In quadrants 3 and 4, MT2 is negative, and current flows from MT1 to MT2, also through P, N, P and N layers. The N region attached to MT2 is active, but the N region attached to MT1 only participates in the initial triggering, not the bulk current flow.

In most applications, the gate current comes from MT2, so quadrants 1 and 3 are the only operating modes (both gate and MT2 positive or negative against MT1). Other applications with single polarity triggering from an IC or digital drive circuit operate in quadrants 2 and 3, than MT1 is usually connected to positive voltage (e.g. +5V) and gate is pulled down to 0V (ground).

3.16 Electrolyte Capacitor

An electrolytic capacitor is a type of capacitor that uses an electrolyte to achieve a larger capacitance than other capacitor types. An electrolyte is a liquid or gel containing a high concentration of ions. Almost all electrolytic capacitors are polarized, which means that the voltage on the positive terminal must always be

greater than the voltage on the negative terminal. The benefit of large capacitance in electrolytic capacitors comes with several drawbacks as well. Among these drawbacks are large leakage currents, value tolerances, equivalent series resistance and a limited lifetime. Electrolytic capacitors can be either wet-electrolyte or solid polymer. They are commonly made of tantalum or aluminum, although other materials may be used. Supercapacitors are a special subtype of electrolytic capacitors, also called double-layer electrolytic capacitors, with capacitances of hundreds and thousands of farads. This article will be based on aluminum electrolytic capacitors. These have a typical capacitance between $1\mu\text{F}$ to 47mF and an operating voltage of up to a few hundred volts DC. Aluminum electrolytic capacitors are found in many applications such as power supplies, computer motherboards and many domestic appliances. Since they are polarized, they may be used only in DC circuits.



Fig.3.21Electrolyte Capacitor

3.16.1 Characteristics

Capacitance drift

The capacitance of electrolytic capacitors drifts from the nominal value as time passes, and they have large tolerances, typically 20%. This means that an aluminum electrolytic capacitor with a nominal capacitance of $47\mu\text{F}$ is expected to have a measured value of anywhere between $37.6\mu\text{F}$ and $56.4\mu\text{F}$. Tantalum electrolytic capacitors can be made with tighter tolerances, but their maximum operating voltage is lower so they cannot be always used as a direct replacement.

Polarity and safety

Due to the construction of electrolytic capacitors and the characteristics of the electrolyte used, electrolytic capacitors must be forward biased. This means that the positive terminal must always be at a higher voltage than the negative terminal. If the capacitor becomes revers-biased (if the voltage polarity on the terminals is reversed),

the insulating aluminum oxide, which acts as a dielectric, might get damaged and start acting as a short circuit between the two capacitor terminals. This can cause the capacitor to overheat due to the large current running through it. As the capacitor overheats, the electrolyte heats up and leaks or even vaporizes, causing the enclosure to burst. This process happens at reverse voltages of about 1 volt and above. To maintain safety and prevent the enclosure from exploding due to high pressures generated under overheat conditions, a safety valve is installed in the enclosure. It is typically made by making a score in the upper face of the capacitor, which pops open in a controlled manner when the capacitor overheats. Since electrolytes may be toxic or corrosive, additional safety measures may need to be taken when cleaning after and replacing an overheated electrolytic capacitor.

There is a special type of electrolytic capacitors for AC use, which is designed to withstand reverse polarisation. This type is called the non-polarized or NP type.

3.17 Zero Crossing Detector Circuit and its Applications

A zero crossing detector or ZCD is a one type of voltage comparator, used to detect a sine waveform transition from positive and negative, that coincides when the i/p crosses the zero voltage condition. In this article, we discuss about zero crossing detector circuit with two different circuits, working principle, theory and applications. The applications of Zero Crossing Detector are phase meter and time marker generator.

A zero-crossing is a point where the sign of a mathematical function changes (e.g. from positive to negative), represented by a intercept of the axis (zero value) in the graph of the function. It is a commonly used term in electronics, mathematics, acoustics, and image processing. In alternating current, the zero-crossing is the instantaneous point at which there is no voltage present. In a sine wave or other simple waveform, this normally occurs twice during each cycle. It is a device for detecting the point where the voltage crosses zero in either direction

The zero-crossing is important for systems which send digital data over AC circuits, such as modems, X10 home automation control systems, and Digital Command Control type systems for Lionel and other AC trains. Counting zero-crossings is also a method used in speech processing to estimate the fundamental frequency of speech.

In a system where an amplifier with digitally controlled gain is applied to an input signal, artifacts in the non-zero output signal occur when the gain of the amplifier is abruptly switched between its discrete gain settings. At audio frequencies, such as in modern consumer electronics like digital audio players, these effects are clearly audible, resulting in a 'zipping' sound when rapidly ramping the gain, or a soft 'click' when a single gain change is made. Artifacts are disconcerting and clearly not desirable. If changes are made only at zero-crossings of the input signal, then no matter how the amplifier gain setting changes, the output also remains at zero, thereby minimizing the change. (The instantaneous change in gain will still produce distortion, but will not produce a click.)

If electrical power is to be switched, no electrical interference is generated if switched at an instant when there is no current—a zero crossing. Early light dimmers and similar devices generated interference; later versions were designed to switch at the zero crossing.

CHAPTER-4

Circuit Diagram

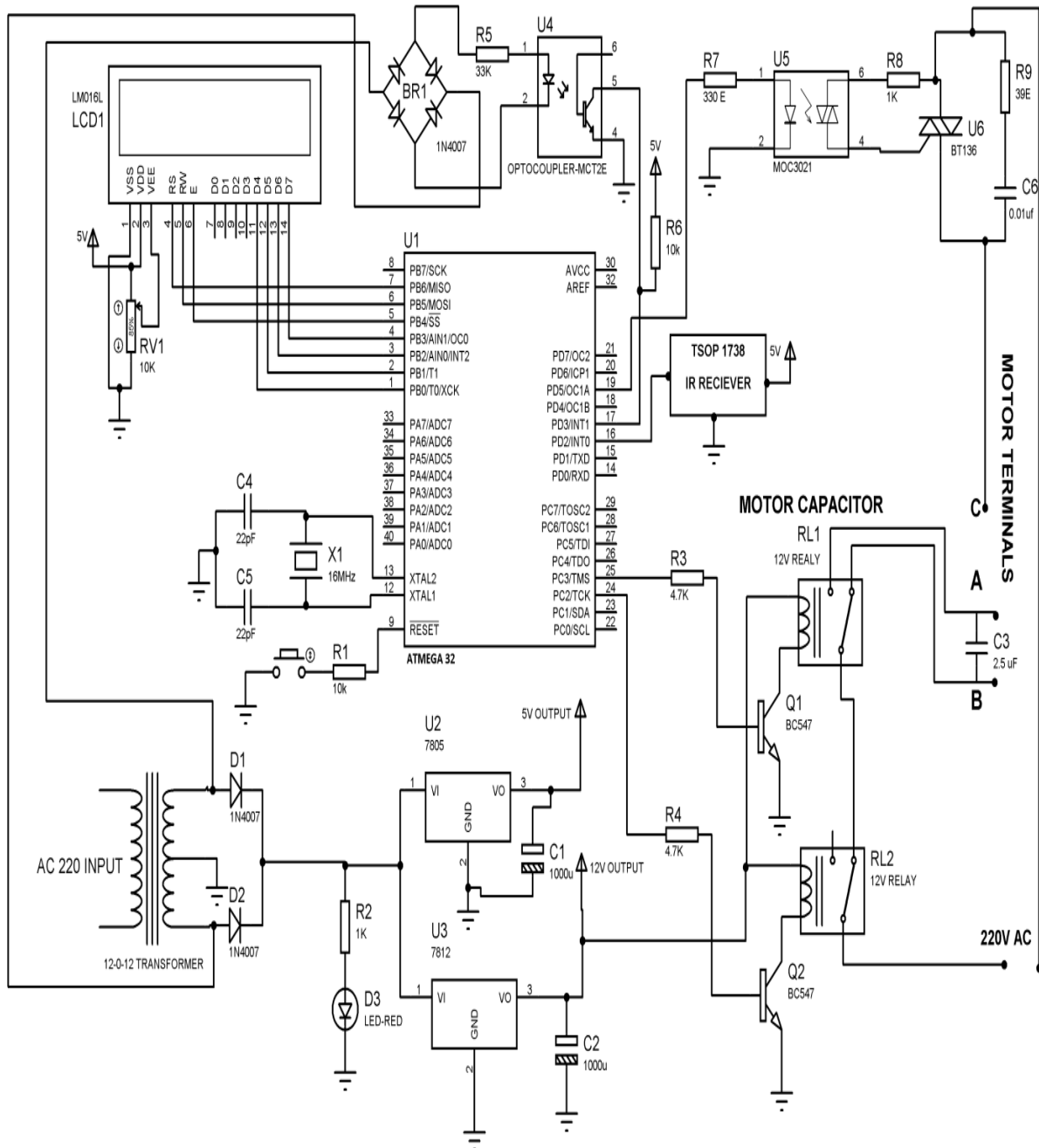


Fig.4.Circuit Diagram

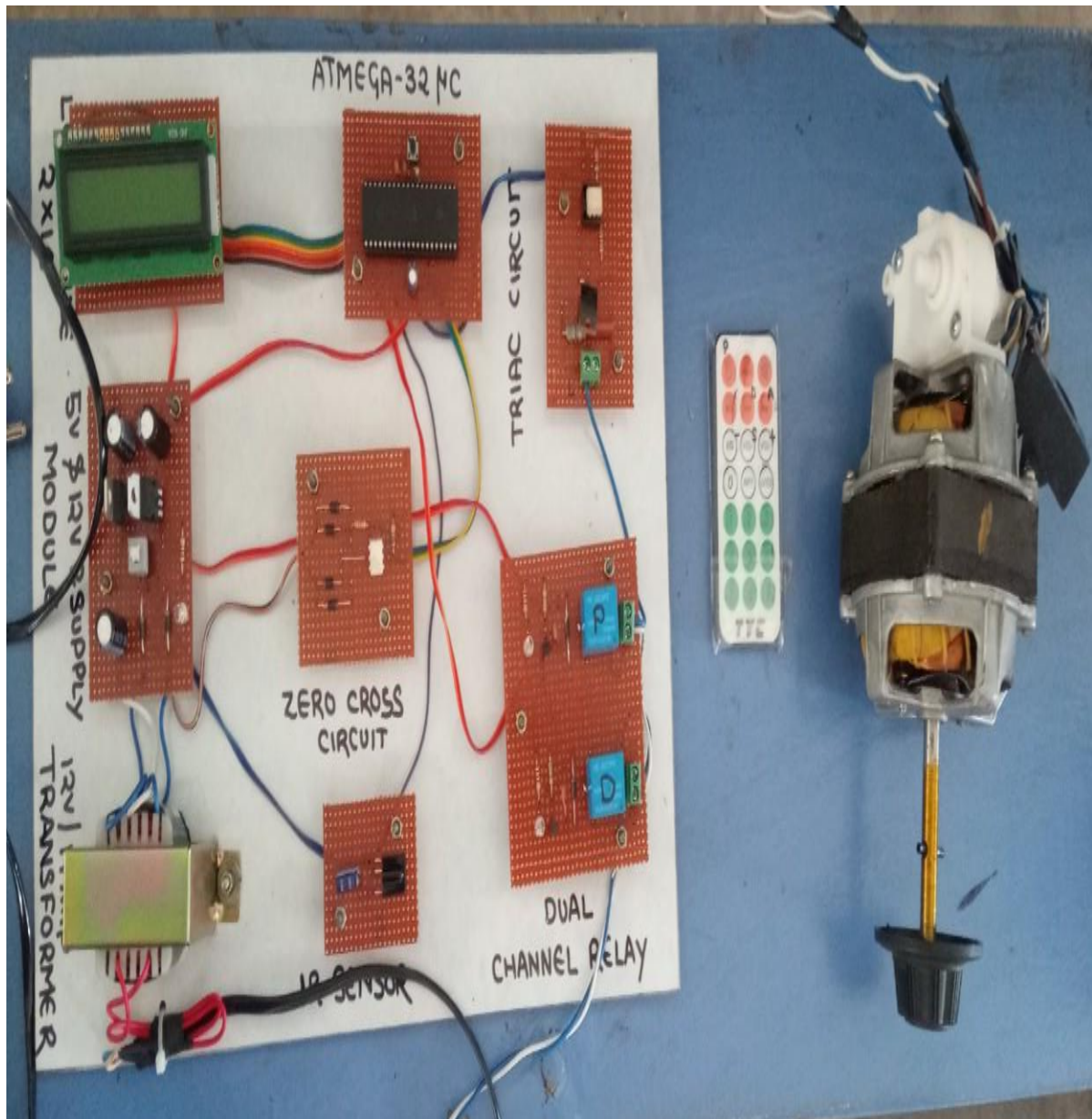
The working of this bidirectional rotation of a DC motor with a remote-control device is very simple. For controlling the direction of rotation of any type DC motor or ac induction motor such as exhaust fan or any motor, this device is connected in series with main supply

of that specific circuit. This device mostly consists of electronic components which are operated at 5 or 6V dc. For making 5 or 6V dc, main supply is stepped down to 12V ac with the help of single phase transformer then these voltages are converted into dc with the help of bridge rectifier. After that, these voltages are regulated into 5V dc with the help of voltage regulator. From these regulated voltages the microcontroller and IR sensors are powered up. Microcontroller is the main controller of this device. IR sensor are basically the infrared sensors which can be operate with any type of IR remote. When IR remote switch is pressed then IR sensors are switched on. When these are switched on, then these give the logic high signal to microcontroller. Then microcontroller efficiently controls the direction of rotation of motor. In this bidirectional rotation of a motor control device, we have configured the remote switch no.1 for forward direction of rotation of motor, switch no.2 for stop and switch no.3 for reverse direction of rotation of motor. This was done only for make the convenient for user interface. Two relays have been used for making the polarities of supply to change the motor rotating direction. These relays are derived through relay driver IC. Further it can extend to ac motors which have wide range of applications. Applications and Advantages of Bidirectional Rotation of a Motor with a Remote Control Device By using this bidirectional rotation of a motor with a remote control device the user can easily rotate the series motor in either direction. This bidirectional rotation of an induction motor device could be easily used with domestic and industrial exhaust fans for fresh air in and hot air throw out. By using bidirectional rotation of an induction motor device, the user can easily on or off the split phase inducing motor with some distance without wasting any time.

4.1 Applications

- Kitchens: While cooking, lots of smoke, heat, odour, and oils are released. When installed in kitchens, exhaust fans help push out these unwanted pollutants from the kitchen, thereby, keeping it fresh and dry.
- Industries: In many industries, smoke, steam, and other air pollutants are released due to the industrial activities. These need to be eliminated from within the buildings to prevent dampness and corrosion of mechanical equipment. A powerful industrial exhaust fan can be ideal for driving these pollutants out and keep the moisture and humidity in check.

Hard ware Model for Bi-directional Speed control of



Induction motor

Fig.5 Bi-directional and speed control of induction motor

The working of this bidirectional rotation of an induction motor with a remote-control device is very simple. For controlling the direction of rotation of any type of single-phase ac induction motor such as exhaust fan or any motor, this device is connected in series with main supply of that specific single-phase motor. This device mostly consists of electronic components which are operated at 5 or 6V dc. For making 5 or 6V dc, main supply is stepped down to 6 or 9V ac with the help of single-phase transformer then these voltages are converted into dc with the help of bridge rectifier. After that, these voltages are regulated into 5V dc with the help of voltage regulator.

From these regulated voltages the pic microcontroller and IR sensors are powered up. Microcontroller is the main controller of this device. It is programmed in c language with the help of mikro/c software. IR sensor are basically the infrared sensors which can be operate with any type of IR remote. When IR remote switch is pressed then IR sensors are switched on. When these are switched on, then these gives the logic high signal to pic microcontroller. Then microcontroller efficiently control the direction of rotation of split phase induction motor. Split phase induction motor is directly powered up with 230V ac, but its control circuit is make on or off through microcontroller. In this bidirectional rotation of an induction motor control device, we have configured the remote switch no.1 for forward direction of rotation of motor, switch no.2 for stop and switch no.3 for reverse direction of rotation of split phase induction motor. This have only done for make the convenient for user interface. Two relays have been used for making on or off the split phase induction motor. These relays are derived through relay driver IC ULN 2803.

CHAPTER-6

CONCLUSION

The proposed system is intended to drive an induction motor for the necessary application in forward and reverse directions using wireless technology. It is observed that when supply is switched on, the motor starts rotating in clockwise direction, and on pressing the push button present on the TV remote enables the motor to rotate in anti-clockwise direction. Using the microcontroller program enables the motor to rotate in both directions (clockwise and anticlockwise direction). It is simpler compared to other methods of rotating motors, flexible in design, lighter in weight and suitable for low power applications. In future this idea can be implemented by using ac motors to control the rotation motor in both directions.

CHAPTER-7

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